Durham University

# Examination Paper

| Examination Session: | Year: | Exam Code: |
|---|---|---|
| May/June | 2023 | ENGI3411-WE01 |

**Title: (BEng/MEng)**

Solid Mechanics 3

| | |
|---|---|
| Time Allowed: | 2 hours |
| Additional Material provided: | Yes - Formula Sheet included as part of exam paper. |
| Materials Permitted: | N/A |
| Calculators Permitted: | Those from the Casio fx-83 and fx-85 series. |

| | |
|---|---|
| **Instructions to Candidates:** | This is an in-person, invigilated exam.<br><br>Answer ALL questions.<br>Use a separate answer book for each question.<br>All relevant workings must be shown. |

| Revision: | |
|---|---|

# Q1

Important: marks in all parts will be awarded not only for your mathematics, but also for the **quality and clarity of your explanations**, including the **discussion of key assumptions**.

(a) A square alloy plate of size 1 m $\times$ 1 m, and of thickness 5 mm, is subjected to a uniformly distributed, in-plane uniaxial tensile force of 330 kN in a direction aligned with one of the edges of the plate. You can easily calculate, then, that the stress applied is 66 MPa. The fracture toughness of the alloy is 30 MPa$\sqrt{\text{m}}$. Determine the critical length of an edge crack that might develop from the mid-point of an edge, and also find the value of the maximum principal stress at a point 10 mm ahead of the crack tip immediately prior to fracture. **[25%]**

(b) The applied load now oscillates such that 330 kN remains the maximum applied force, but there is now a non-zero, positive stress ratio $R$. Using the available equipment, the shortest detectable edge crack is 0.75 mm in length. You are going to analyse the number of load cycles, $N_f$, it takes for a crack that is just detectable to reach critical so that the plate fails in brittle fracture. Since $R > 0$, instead of the Paris law, you are asked to use a Walker crack growth law of the form

$$\frac{da}{dN} = \frac{C(\Delta K)^m}{(1-R)^n}$$

where the symbols have their usual meanings, and (assuming the crack growth rate and stress intensity factors are in units of m/cycle and MPa$\sqrt{\text{m}}$ respectively) you may use the following material coefficients for the alloy: $C = 1.5 \times 10^{-10}$, $m = 3.2$, $n = 1.7$.

Develop an expression for $N_f$ in terms of the unknown stress ratio, $R$. **[30%]**

(c) Laboratory testing shows you that, for the situation considered in part (b), the number of cycles $N_f$ is 200,000. Determine the stress ratio $R$ and thereby determine the minimum force applied to the plate over each loading cycle. **[10%]**

(d) An SN curve for the alloy (produced using tests at $R = -1$) is presented in Figure Q1. Estimate the fatigue life for the plate (i.e. cycles to failure) when loaded with the same stress amplitude as considered in part (b) above, but with a stress ratio $R = -1$. In your own words, explain any discrepancy between this fatigue life and the 200,000 cycles to failure given in part (c). **[25%]**

(e) An alternative to the Paris and Walker crack growth laws is that due to Forman:

$$\frac{da}{dN} = \frac{C(\Delta K)^m}{(1-R)(K_{Ic} - K_{max})}$$

What do you think is the reason for including the term $(K_{Ic} - K_{max})$ in the denominator? **[10%]**

**continued**

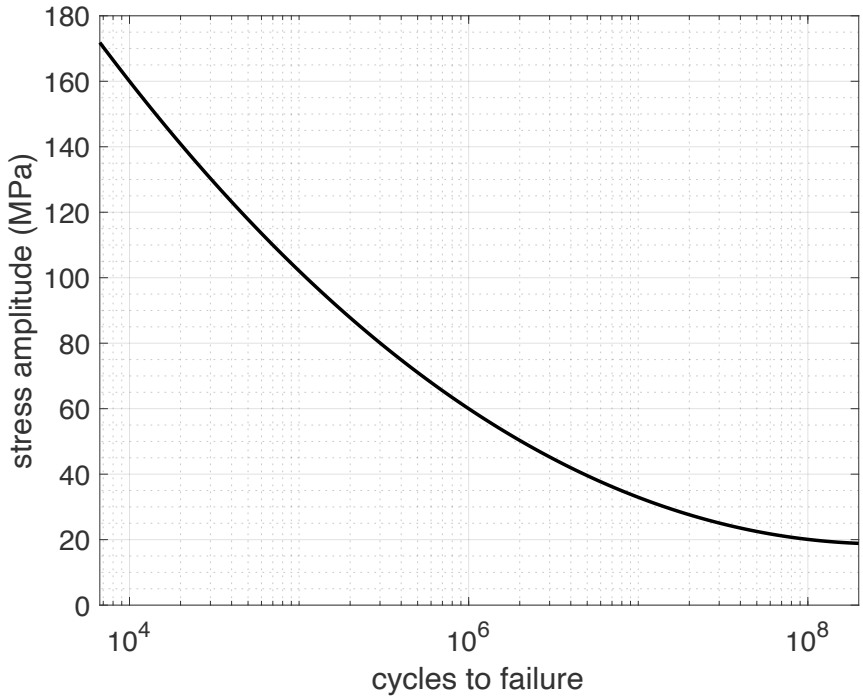



Figure Q1: SN curve for alloy

**continued**

## Q2

When analysing components using the finite element method we often want to grade the mesh that we are using so that we can concentrate the degrees of freedom (i.e. the nodes and elements) in the areas of interest, such as in areas where we expect to have stress concentrations due to changes in geometry. We may also wish to use different elements in different parts of the mesh, which can require the use of transition elements. This question is focused on such elements.

All finite elements should satisfy completeness and compatibility requirements.

(a) In your own words, explain the implications of the compatibility requirement on the displacement variation over a stress analysis problem. **[10%]**

Consider the finite element mesh shown in Figure Q2. The left of the figure shows the global finite element mesh which comprises both quadrilateral and triangular elements. Some of the triangular elements, such as the one shaded grey, need to transition between bi-quadratic quadrilaterals and linear triangles, whilst maintaining the compatibility requirement. The local system for this element is shown on the right of the figure.

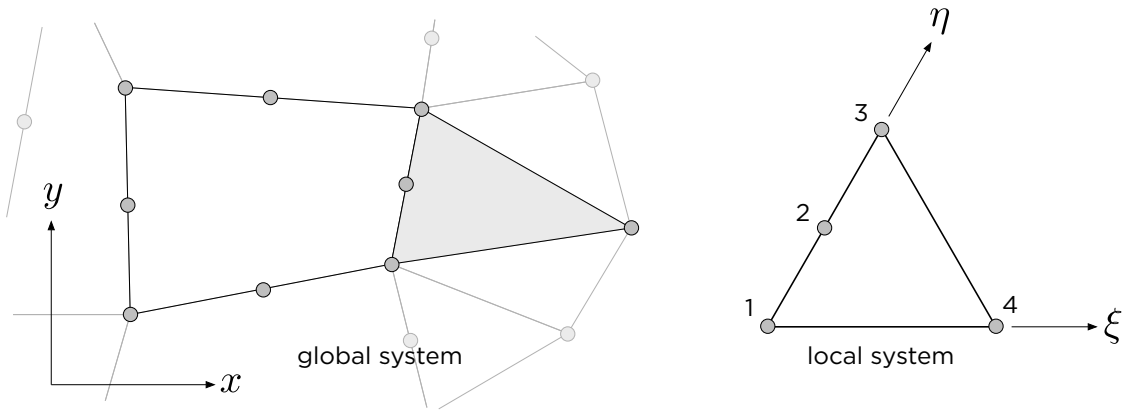



Figure Q2: Global finite element mesh and local element system where $\xi, \eta \in [0, 1]$.

(b) Assuming that the shape functions of the triangular element shown in Figure Q2 have the form

$$N_i = a_i + b_i\xi + c_i\eta + d_i\eta^2,$$

show that the elements shape functions are

$$N_1 = 1 - \xi - 3\eta + 2\eta^2, \qquad N_2 = 4\eta(1 - \eta), \qquad N_3 = \eta(2\eta - 1) \quad \text{and} \quad N_4 = \xi.$$

Include all of your working and perform the appropriate checks. **[65%]**

(c) If the displacement vector of the triangular element is

$$\{d\} = \{0.05 \quad 0.03 \quad 0.12 \quad 0.03 \quad 0.09 \quad 0.03 \quad -0.02 \quad 0.03\}^T \text{ m},$$

where the entries are ordered according to the local convention, sketch the variation of the $x$ displacement over the edge of the element where $\xi = 0$ and $\eta \in [0, 1]$. **[25%]**

**continued**

## Q3

Finite element analysis requires the evaluation of integrals over the surface and within the volume of elements in order to calculate quantities within the weak form statement of equilibrium

$$\int_V [B]^T [D^e][B] dV \ \{d\} = \int_S [N]^T \{t\} dS + \int_V [N]^T \{f^b\} dV.$$

The *de facto* choice when performing this integration is numerical Gauss-Legendre quadrature.

(a) Explain in your own words why most finite element codes use Gauss-Legendre quadrature to evaluate the integrals in the above weak form statement. **[10%]**

(b) Use Gauss-Legendre quadrature to integrate the following function

$$f = 3x^4 + x - 8y^2 + 4y$$

over a four-noded quadrilateral defined by the following nodal coordinates

$$\{x\} = \{0.28 \quad 0.42 \quad 1.12 \quad 0.98\}^T \text{ m} \qquad \text{and} \qquad \{y\} = \{0.67 \quad 1.71 \quad 1.97 \quad 0.93\}^T \text{ m}.$$

The local and global coordinates of this element are linked by

$$x = 0.70 + 0.35\xi + 0.07\eta \qquad \text{and} \qquad y = 1.32 + 0.13\xi + 0.52\eta.$$

Justify the order of the quadrature scheme that you adopt to answer this question **[55%]**

Consider the notched plate shown in Figure Q3. The plate is loaded under tension via the four holes, which are subjected to an imposed displacement via loading pins (not shown). You have been asked to analyse the plate using the finite element method to aid the understanding of the experimental results.

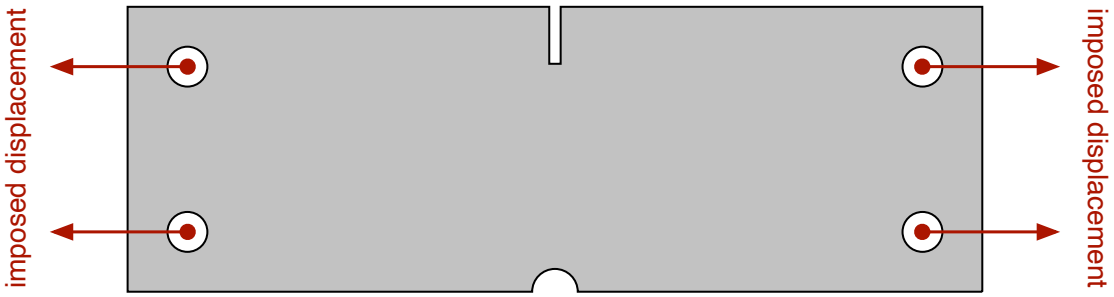



Figure Q3: notched plate under tensile loading

(c) Explain with the aid of a sketch the boundary conditions that you would impose on the plate when analysing it using the finite element method. Explain the assumption you would make in the out-of-plane direction. **[20%]**

(d) If you only had a finite number of elements to analyse the problem, explain how you would distribute the elements within the analysis. Use a sketch to support your answer. **[15%]**

**ENGI3411-WE01 Solid Mechanics 3**
**Course: Fracture and Fatigue**
**Formula sheet**

**Stress and displacement distribution locally around the tip of a mode I crack oriented in the $x$-direction**

$$\sigma_{xx} = \frac{K_I}{\sqrt{2\pi r}} \cos\frac{\theta}{2}\left[1 - \sin\frac{\theta}{2}\sin\frac{3\theta}{2}\right] + ...$$

$$\sigma_{yy} = \frac{K_I}{\sqrt{2\pi r}} \cos\frac{\theta}{2}\left[1 + \sin\frac{\theta}{2}\sin\frac{3\theta}{2}\right] + ...$$

$$\sigma_{xy} = \frac{K_I}{\sqrt{2\pi r}} \sin\frac{\theta}{2}\cos\frac{\theta}{2}\cos\frac{3\theta}{2} + ...$$

$$u_x = \frac{K_\mathrm{I}}{\mu}\sqrt{\frac{r}{2\pi}}\cos\frac{\theta}{2}\left[\frac{1}{2}(\kappa - 1) + \sin^2\frac{\theta}{2}\right] + \cdots$$

$$u_y = \frac{K_\mathrm{I}}{\mu}\sqrt{\frac{r}{2\pi}}\sin\frac{\theta}{2}\left[\frac{1}{2}(\kappa + 1) - \cos^2\frac{\theta}{2}\right] + \cdots$$

$\kappa = 3 - 4\nu$ (plane strain), or $(3-\nu)/(1+\nu)$ (plane stress)

**Mode I stress intensity factor**

$$K_I = Y\sigma\sqrt{\pi a}$$

Internal crack of length $2a$ in infinite plate: $Y = 1$
Edge crack of length $a$ in relatively large plate: $Y = 1.12$

**Plastic zone correction**

$$\text{Equivalent crack length} = a + \frac{K_I^2}{2\pi\sigma_Y^2}$$

**Failure diagrams for mean stress correction**

Modified Goodman:

$$S_a = S_D\left[1 - \frac{S_{mean}}{S_u}\right]$$

Gerber parabola:

$$S_a = S_D\left[1 - \left(\frac{S_{mean}}{S_u}\right)^2\right]$$

Soderberg:

$$S_a = S_D\left[1 - \frac{S_{mean}}{S_{yield}}\right]$$

**Strain-life method**

Strain life equation:

$$\frac{\Delta\varepsilon}{2} = \frac{\sigma_f'}{E}(2N_f)^b + \varepsilon_f'(2N_f)^c$$

Mean stress correction (Smith Watson Topper):

$$\left(\sigma_{mean} + \frac{\Delta\sigma}{2}\right)\frac{\Delta\varepsilon}{2} = \frac{\sigma_f'^2}{E}(2N_f)^{2b} + \sigma_f'\varepsilon_f'(2N_f)^{b+c}$$

**continued**

**Macro crack growth laws based on fracture mechanics approach**

Paris Law:

$$\frac{da}{dN} = C(\Delta K)^m$$

Broek and Schijve:

$$\frac{da}{dN} = CK_{max}^2 \Delta K$$

Walker:

$$\frac{da}{dN} = CK_{max}^m (\Delta K)^n$$

Walker:

$$\frac{da}{dN} = C\overline{\Delta K}^n \qquad \text{where } \overline{\Delta K} = S_{max}(1-R)^m\sqrt{\pi a}$$

Forman:

$$\frac{da}{dN} = \frac{C(\Delta K)^m}{(1-R)(K_{Ic} - K_{max})}$$

**continued**

**ENGI3411-WE01 Solid Mechanics 3**
**Course: Finite-element method**
**Formula sheet**

Three-dimensional stress-strain relationship

$$\begin{Bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \sigma_{zz} \\ \sigma_{xy} \\ \sigma_{yz} \\ \sigma_{zx} \end{Bmatrix} = \frac{E}{(1+\nu)(1-2\nu)} \begin{bmatrix} (1-\nu) & \nu & \nu & 0 & 0 & 0 \\ \nu & (1-\nu) & \nu & 0 & 0 & 0 \\ \nu & \nu & (1-\nu) & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{(1-2\nu)}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{(1-2\nu)}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{(1-2\nu)}{2} \end{bmatrix} \begin{Bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \varepsilon_{zz} \\ \gamma_{xy} \\ \gamma_{yz} \\ \gamma_{zx} \end{Bmatrix}$$

Two-dimensional stress-strain relationships

$$\begin{Bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \sigma_{xy} \end{Bmatrix} = \frac{E}{(1+\nu)(1-2\nu)} \begin{bmatrix} (1-\nu) & \nu & 0 \\ \nu & (1-\nu) & 0 \\ 0 & 0 & \frac{(1-2\nu)}{2} \end{bmatrix} \begin{Bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \gamma_{xy} \end{Bmatrix} \qquad \text{(plane strain)}$$

$$\begin{Bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \sigma_{xy} \end{Bmatrix} = \frac{E}{1-\nu^2} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{(1-\nu)}{2} \end{bmatrix} \begin{Bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \gamma_{xy} \end{Bmatrix} \qquad \text{(plane stress)}$$

Three-dimensional strain definition

$$\begin{Bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \varepsilon_{zz} \\ \gamma_{xy} \\ \gamma_{yz} \\ \gamma_{zx} \end{Bmatrix} = [L]\{u\} = \begin{bmatrix} \frac{\partial}{\partial x} & 0 & 0 \\ 0 & \frac{\partial}{\partial y} & 0 \\ 0 & 0 & \frac{\partial}{\partial z} \\ \frac{\partial}{\partial y} & \frac{\partial}{\partial x} & 0 \\ 0 & \frac{\partial}{\partial z} & \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} & 0 & \frac{\partial}{\partial x} \end{bmatrix} \begin{Bmatrix} u \\ v \\ w \end{Bmatrix}$$

Interpolation using shape functions

$$u = \sum_{i=1}^{n_{en}} N_i u_i = \{N\}^T \{u\}$$

Three-dimensional strain-displacement definition

$$\begin{Bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \varepsilon_{zz} \\ \gamma_{xy} \\ \gamma_{yz} \\ \gamma_{zx} \end{Bmatrix} = \underbrace{\begin{bmatrix} \frac{\partial N_1}{\partial x} & 0 & 0 & \dots & \frac{\partial N_{n_{en}}}{\partial x} & 0 & 0 \\ 0 & \frac{\partial N_1}{\partial y} & 0 & \dots & 0 & \frac{\partial N_{n_{en}}}{\partial y} & 0 \\ 0 & 0 & \frac{\partial N_1}{\partial z} & \dots & 0 & 0 & \frac{\partial N_{n_{en}}}{\partial z} \\ \frac{\partial N_1}{\partial y} & \frac{\partial N_1}{\partial x} & 0 & \dots & \frac{\partial N_{n_{en}}}{\partial y} & \frac{\partial N_{n_{en}}}{\partial x} & 0 \\ 0 & \frac{\partial N_1}{\partial z} & \frac{\partial N_1}{\partial y} & \dots & 0 & \frac{\partial N_{n_{en}}}{\partial z} & \frac{\partial N_{n_{en}}}{\partial y} \\ \frac{\partial N_1}{\partial z} & 0 & \frac{\partial N_1}{\partial x} & \dots & \frac{\partial N_{n_{en}}}{\partial z} & 0 & \frac{\partial N_{n_{en}}}{\partial x} \end{bmatrix}}_{[B]=[L][N]} \begin{Bmatrix} u_1 \\ v_1 \\ w_1 \\ \vdots \\ u_{n_{en}} \\ v_{n_{en}} \\ w_{n_{en}} \end{Bmatrix}$$

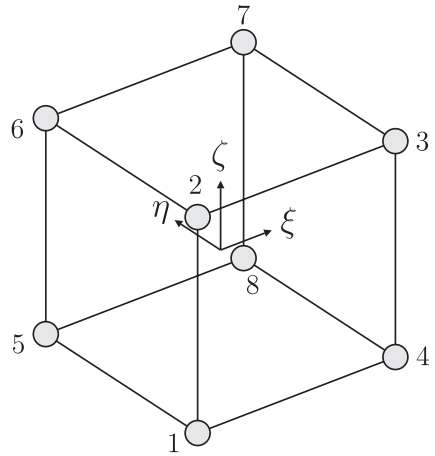
**continued**

Plane-stress/strain strain-displacement definition

$$\left\{\begin{array}{c}\varepsilon_{xx}\\ \varepsilon_{yy}\\ \gamma_{xy}\end{array}\right\} = \left[\begin{array}{cccc}\frac{\partial N_1}{\partial x} & 0 & \cdots & \frac{\partial N_{n_{en}}}{\partial x} & 0\\ 0 & \frac{\partial N_1}{\partial y} & \cdots & 0 & \frac{\partial N_{n_{en}}}{\partial y}\\ \frac{\partial N_1}{\partial y} & \frac{\partial N_1}{\partial x} & \cdots & \frac{\partial N_{n_{en}}}{\partial y} & \frac{\partial N_{n_{en}}}{\partial x}\end{array}\right] \left\{\begin{array}{c}u_1\\ v_1\\ \vdots\\ u_{n_{en}}\\ v_{n_{en}}\end{array}\right\}.$$

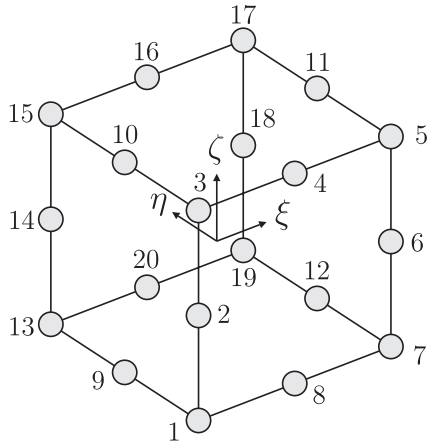
Shape functions for hexahedral elements

Tri-linear eight-noded hexahedral




$$\begin{aligned}
N_1 &= (1-\xi)(1-\eta)(1-\zeta)/8\\
N_2 &= (1-\xi)(1-\eta)(1+\zeta)/8\\
N_3 &= (1+\xi)(1-\eta)(1+\zeta)/8\\
N_4 &= (1+\xi)(1-\eta)(1-\zeta)/8\\
N_5 &= (1-\xi)(1+\eta)(1-\zeta)/8\\
N_6 &= (1-\xi)(1+\eta)(1+\zeta)/8\\
N_7 &= (1+\xi)(1+\eta)(1+\zeta)/8\\
N_8 &= (1+\xi)(1+\eta)(1-\zeta)/8
\end{aligned}$$

Tri-quadratic twenty-noded hexahedral




$$\begin{aligned}
N_1 &= (1-\xi)(1-\eta)(1-\zeta)(-\xi-\eta-\zeta-2)/8\\
N_2 &= (1-\xi)(1-\eta)(1-\zeta^2)/4\\
N_3 &= (1-\xi)(1-\eta)(1+\zeta)(-\xi-\eta+\zeta-2)/8\\
N_4 &= (1-\xi^2)(1-\eta)(1+\zeta)/4\\
N_5 &= (1+\xi)(1-\eta)(1+\zeta)(\xi-\eta+\zeta-2)/8\\
N_6 &= (1+\xi)(1-\eta)(1-\zeta^2)/4\\
N_7 &= (1+\xi)(1-\eta)(1-\zeta)(\xi-\eta-\zeta-2)/8\\
N_8 &= (1-\xi^2)(1-\eta)(1-\zeta)/4\\
N_9 &= (1-\xi)(1-\eta^2)(1-\zeta)/4\\
N_{10} &= (1-\xi)(1-\eta^2)(1+\zeta)/4\\
N_{11} &= (1+\xi)(1-\eta^2)(1+\zeta)/4\\
N_{12} &= (1+\xi)(1-\eta^2)(1-\zeta)/4\\
N_{13} &= (1-\xi)(1+\eta)(1-\zeta)(-\xi+\eta-\zeta-2)/8\\
N_{14} &= (1-\xi)(1+\eta)(1-\zeta^2)/4\\
N_{15} &= (1-\xi)(1+\eta)(1+\zeta)(-\xi+\eta+\zeta-2)/8\\
N_{16} &= (1-\xi^2)(1+\eta)(1+\zeta)/4\\
N_{17} &= (1+\xi)(1+\eta)(1+\zeta)(\xi+\eta+\zeta-2)/8\\
N_{18} &= (1+\xi)(1+\eta)(1-\zeta^2)/4\\
N_{19} &= (1+\xi)(1+\eta)(1-\zeta)(\xi+\eta-\zeta-2)/8\\
N_{20} &= (1-\xi^2)(1+\eta)(1-\zeta)/4
\end{aligned}$$

**continued**

Axi-symmetric strain-displacement definition

$$\begin{Bmatrix} \varepsilon_{rr} \\ \varepsilon_{zz} \\ \varepsilon_{\theta\theta} \\ \gamma_{rz} \end{Bmatrix} = \begin{bmatrix} \frac{\partial N_1}{\partial r} & 0 & \cdots & \frac{\partial N_{nen}}{\partial r} & 0 \\ 0 & \frac{\partial N_1}{\partial z} & \cdots & 0 & \frac{\partial N_{nen}}{\partial z} \\ \frac{N_1}{r_1} & 0 & \cdots & \frac{N_{nen}}{r_{nen}} & 0 \\ \frac{\partial N_1}{\partial z} & \frac{\partial N_1}{\partial r} & \cdots & \frac{\partial N_{nen}}{\partial z} & \frac{\partial N_{nen}}{\partial r} \end{bmatrix} \begin{Bmatrix} u_1 \\ w_1 \\ \vdots \\ u_{n_{en}} \\ w_{n_{en}} \end{Bmatrix}.$$

Global shape function derivatives

$$\begin{Bmatrix} \frac{\partial N_i}{\partial x} \\ \frac{\partial N_i}{\partial y} \\ \frac{\partial N_i}{\partial z} \end{Bmatrix} = \begin{bmatrix} \frac{\partial \xi}{\partial x} & \frac{\partial \eta}{\partial x} & \frac{\partial \zeta}{\partial x} \\ \frac{\partial \xi}{\partial y} & \frac{\partial \eta}{\partial y} & \frac{\partial \zeta}{\partial y} \\ \frac{\partial \xi}{\partial z} & \frac{\partial \eta}{\partial z} & \frac{\partial \zeta}{\partial z} \end{bmatrix} \begin{Bmatrix} \frac{\partial N_i}{\partial \xi} \\ \frac{\partial N_i}{\partial \eta} \\ \frac{\partial N_i}{\partial \zeta} \end{Bmatrix}.$$

The Jacobian matrix

$$[J] = \begin{bmatrix} \frac{\partial x}{\partial \xi} & \frac{\partial y}{\partial \xi} & \frac{\partial z}{\partial \xi} \\ \frac{\partial x}{\partial \eta} & \frac{\partial y}{\partial \eta} & \frac{\partial z}{\partial \eta} \\ \frac{\partial x}{\partial \zeta} & \frac{\partial y}{\partial \zeta} & \frac{\partial z}{\partial \zeta} \end{bmatrix} = \begin{bmatrix} \frac{\partial N_1}{\partial \xi} & \frac{\partial N_2}{\partial \xi} & \cdots & \frac{\partial N_{n_{en}}}{\partial \xi} \\ \frac{\partial N_1}{\partial \eta} & \frac{\partial N_2}{\partial \eta} & \cdots & \frac{\partial N_{n_{en}}}{\partial \eta} \\ \frac{\partial N_1}{\partial \zeta} & \frac{\partial N_2}{\partial \zeta} & \cdots & \frac{\partial N_{n_{en}}}{\partial \zeta} \end{bmatrix} \begin{bmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ \vdots & \vdots & \vdots \\ x_{n_{en}} & y_{n_{en}} & z_{n_{en}} \end{bmatrix}$$

Weak form of equilibrium

$$\int_V [B]^T [D^{\mathsf{e}}][B] dV \{d\} = \int_S [N]^T \{t\} dS + \int_V [N]^T \{f^b\} dV.$$

Element stiffness and force vector

$$[k^e] = \int_V [B]^T [D^{\mathsf{e}}][B] dV \qquad \text{and} \qquad \{f^e\} = \int_S [N]^T \{t\} dS + \int_V [N]^T \{f^b\} dV.$$

Gauss-Legendre quadrature

$$\int_{-1}^{1} f(x) dx \approx \sum_{i=1}^{n_{Gp}} f(x_i) w_i$$

| $n_{Gp}$ | Abscissae, $x_i$ | Weights, $w_i$ |
|---|---|---|
| 1 | $0$ | $0$ |
| 2 | $\pm 1/\sqrt{3}$ | $1$ |
| 3 | $0$ | $8/9$ |
| | $\pm\sqrt{3/5}$ | $5/9$ |
| 4 | $\pm\sqrt{\frac{3}{7} - \frac{2}{7}\sqrt{\frac{6}{5}}}$ | $(18+\sqrt{30})/36$ |
| | $\pm\sqrt{\frac{3}{7} + \frac{2}{7}\sqrt{\frac{6}{5}}}$ | $(18-\sqrt{30})/36$ |

Numerical approximation of the element stiffness matrix

$$[k^e] \approx \sum_{i=1}^{n_{Gp}} [B]_i^T [D^{\mathsf{e}}][B]_i w_i \det([J]_i)$$

**END**